

Intro to Transformations – Worksheet

MCR3U

Jensen

1) Describe the transformations, in order, that are being done to the function $f(x)$.

a) $g(x) = -4f(x)$

b) $g(x) = f(3x)$

c) $g(x) = \frac{1}{2}f(-x)$

d) $g(x) = -\frac{1}{3}f\left[\frac{1}{2}(x + 1)\right]$

e) $g(x) = 5f[-2(x - 4)]$

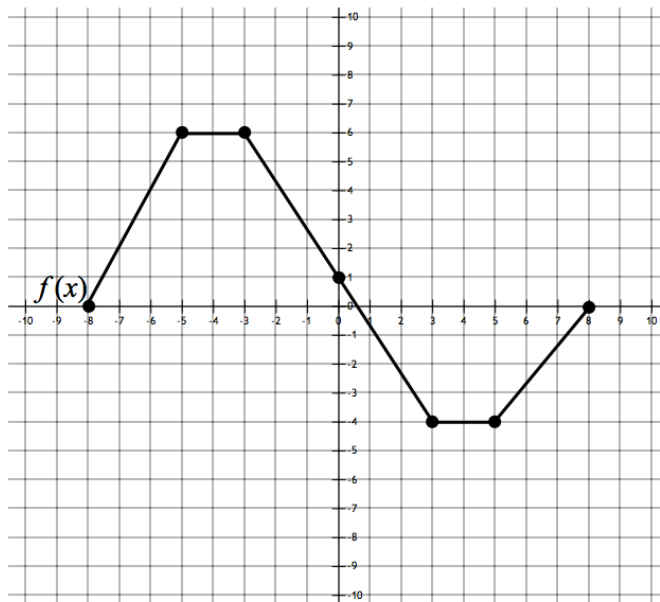
f) $g(x) = -2f(8x) + 4$

h) $g(x) = -\frac{1}{4}f[-3(x - 1)] - 5$

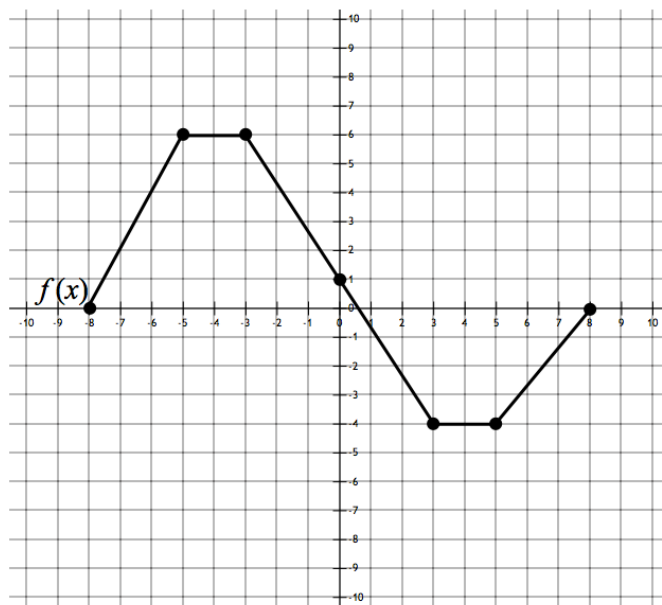
i) $g(x) = 4f\left[-\frac{1}{2}(x + 2)\right] - 1$

2) For the graph of $f(x)$ given, sketch the graph of $g(x)$ after the given transformation.

a) $g(x) = 2f(x) - 2$



b) $g(x) = \frac{1}{2}f(x-1) + 1$



Answers

1) a) vertical reflection over the x-axis and vertical stretch bafo 4 ($-4y$)

b) horizontal compression bafo $\frac{1}{3}$ ($\frac{x}{3}$)

c) vertical compression bafo $\frac{1}{2}$ ($\frac{y}{2}$), horizontal relection over the y-axis ($-x$)

d) vertical reflection over the x-axis and vertical compression bafo $\frac{1}{3}$ ($\frac{y}{-3}$), horizontal stretch bafo 2 ($2x$), phase shift left 1 unit ($x - 1$)

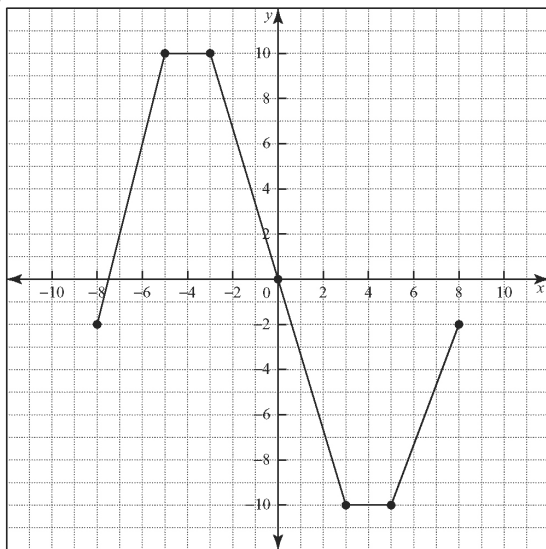
e) vertical stretch bafo 5 ($5y$), horizontal reflection over the y-axis and horizontal compression bafo $\frac{1}{2}$ ($\frac{x}{-2}$), phase shift right 4 units ($x + 4$)

f) vertical reflection over the x-axis and vertical stretch bafo 2 ($-2y$), horizontal compression bafo $\frac{1}{8}$ ($\frac{x}{8}$), shift up 4 units ($y + 4$)

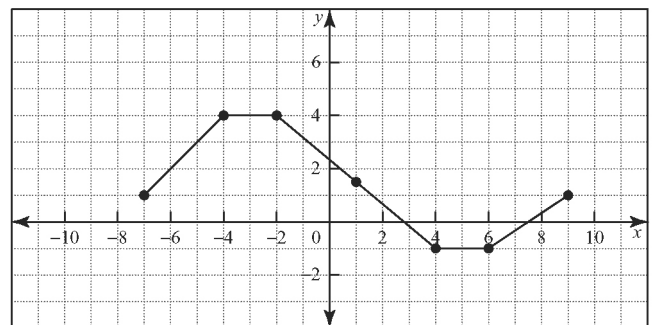
h) vertical reflection over the x-axis and vertical compression bafo $\frac{1}{4}$ ($\frac{y}{-4}$), horizontal reflection over the y-axis and horizontal compression bafo $\frac{1}{3}$ ($\frac{x}{-3}$), phase shift right 1 unit ($x + 1$), shift down 5 units ($y - 5$)

i) vertical stretch bafo 4 ($4y$), horizontal reflection over the y-axis and horizontal stretch bafo 2 ($-2x$), phase shift left 2 units ($x - 2$), shift down 1 unit ($y - 1$)

2) a)



b)



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SOLUTIONS

1) Describe the transformations, in order, that are being done to the function $f(x)$.

a) $g(x) = -4f(x)$

- vertical reflection over the x-axis $(-y)$
- vertical stretch b.o.f.o 4 $(4y)$

b) $g(x) = f(3x)$

- horizontal compression b.o.f.o $\frac{1}{3}$ $(\frac{x}{3})$

c) $g(x) = \frac{1}{2}f(-x)$

- vertical compression b.o.f.o $\frac{1}{2}$ $(\frac{y}{2})$
- horizontal reflection over the y-axis $(-x)$

d) $g(x) = -\frac{1}{3}f[\frac{1}{2}(x+1)]$

- vertical compression b.o.f.o $\frac{1}{3}$ $(\frac{y}{3})$
- vertical reflection over the x-axis $(-y)$
- horizontal stretch b.o.f.o 2 $(2x)$
- phase shift 1 unit left $(x-1)$

e) $g(x) = 5f[-2(x-4)]$

- vertical stretch b.o.f.o 5 $(5y)$
- horizontal compression b.o.f.o $\frac{1}{2}$ $(\frac{x}{2})$
- horizontal reflection across the y-axis $(-x)$
- phase shift 4 units right $(x+4)$

f) $g(x) = -2f(8x) + 4$

- vertical stretch b.o.f.o 2 $(2y)$
- vertical reflection over the x-axis $(-y)$
- horizontal compression b.o.f.o $\frac{1}{8}$ $(\frac{x}{8})$
- shift up 4 units $(y+4)$

h) $g(x) = -\frac{1}{4}f[-3(x-1)] - 5$

- vertical compression b.o.f.o $\frac{1}{4}$ $(\frac{y}{4})$
- vertical reflection over x-axis $(-y)$
- horizontal compression b.o.f.o $\frac{1}{3}$ $(\frac{x}{3})$
- horizontal reflection over y-axis $(-x)$
- phase shift 1 unit right $(x+1)$
- shift 5 units down $(y-5)$

i) $g(x) = 4f[-\frac{1}{2}(x+2)] - 1$

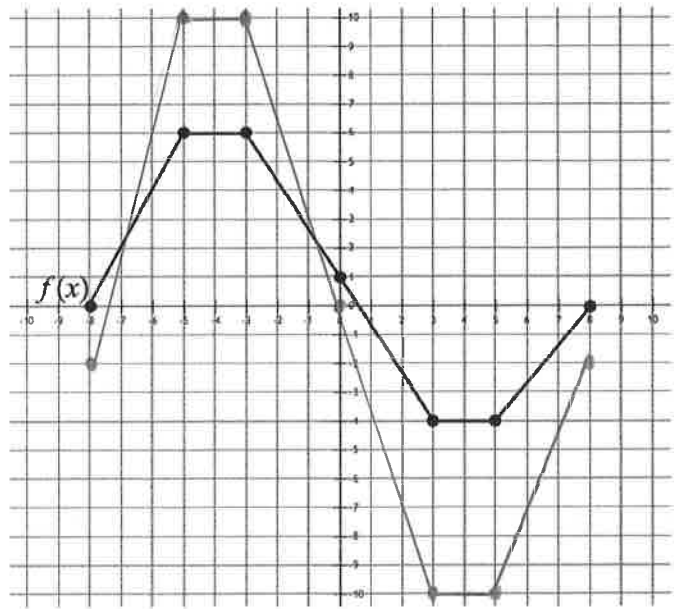
- vertical stretch b.o.f.o 4 $(4y)$
- horizontal stretch b.o.f.o 2 $(2x)$
- horizontal reflection over y-axis $(-x)$
- phase shift 2 units left $(x-2)$
- shift down 1 unit $(y-1)$

2) For the graph of $f(x)$ given, sketch the graph of $g(x)$ after the given transformation.

a) $g(x) = 2f(x) - 2$

- vertical stretch base 2 ($2y$)
- shift down 2 units ($y-2$)

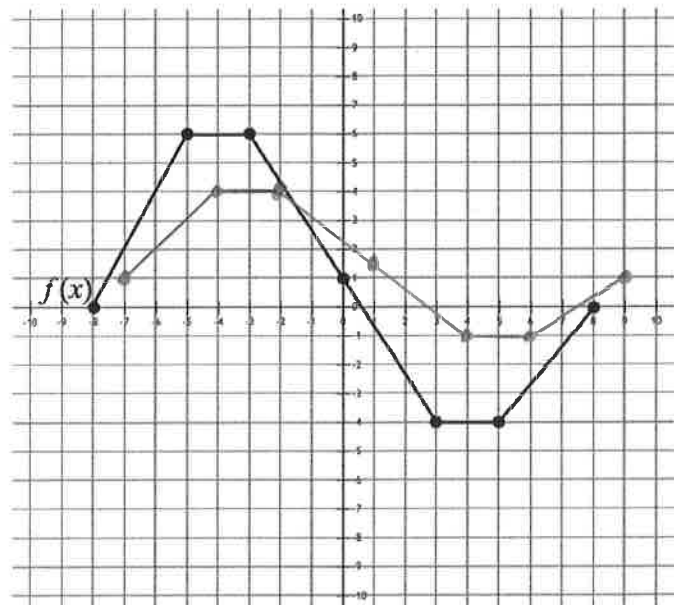
$$(x, 2y-2)$$



b) $g(x) = \frac{1}{2}f(x-1)+1$

- vertical compression base $\frac{1}{2}$ ($\frac{y}{2}$)
- phase shift right 1 unit ($x+1$)
- phase shift up 1 unit ($y+1$)

$$(x+1, \frac{y}{2}+1)$$



Answers

1) a) vertical reflection over the x-axis and vertical stretch bafo 4 ($-4y$)

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c) vertical compression bafo $\frac{1}{2}$ ($\frac{y}{2}$), horizontal relection over the y-axis ($-x$)

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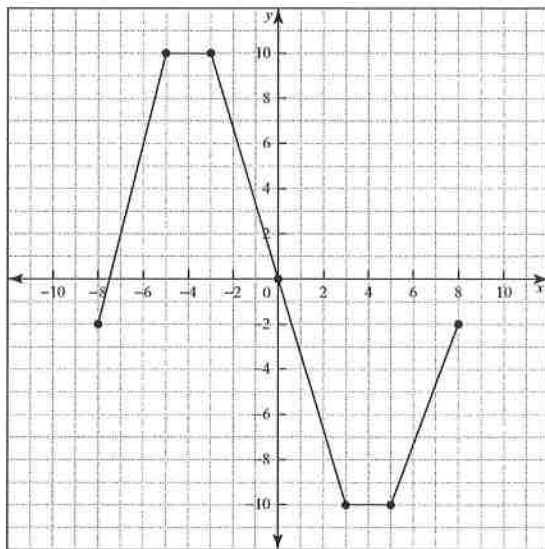
e) vertical stretch bafo 5 ($5y$), horizontal reflection over the y-axis and horizontal compression bafo $\frac{1}{2}$ ($\frac{x}{-2}$), phase shift right 4 units ($x + 4$)

f) vertical reflection over the x-axis and vertical stretch bafo 2 ($-2y$), horizontal compression bafo $\frac{1}{8}$ ($\frac{x}{8}$), shift up 4 units ($y + 4$)

h) vertical reflection over the x-axis and vertical compression bafo $\frac{1}{4}$ ($\frac{y}{-4}$), horizontal reflection over the y-axis and horizontal compression bafo $\frac{1}{3}$ ($\frac{x}{-3}$), phase shift right 1 unit ($x + 1$), shift down 5 units ($y - 5$)

i) vertical stretch bafo 4 ($4y$), horizontal reflection over the y-axis and horizontal stretch bafo 2 ($-2x$), hase shift left 2 units ($x - 2$), shift down 1 unit ($y - 1$)

2) a)



b)

